

An introduction to Python language

Constructor
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Define a functions

built-in function

```
from math import sqrt  
print(sqrt(100))
```

Functions

- *Function blocks begin with the keyword **def** followed by the function name and parentheses (()).
- *Any input parameters or arguments should be placed within these parentheses. You can also define parameters inside these parentheses.
- *The first statement of a function can be an optional statement - the documentation string of the function or *docstring*.
- *The code block within every function starts with a colon (:) and is indented.
- *The statement **return** [expression] exits a function, optionally passing back an expression to the caller. A return statement with no arguments is the same as **return None**.

Define a functions

```
def functionname (parameter):
```

 block of code

```
    return (values)
```

← Optional

Example without returning a value

```
def hello ():
```

```
    print("hello world")
```

التصريح عن الدالة

```
hello()
```

←

استدعاء الدالة

Output

```
"hello world"
```

Examples on functions

Example with returning a value

```
def getstring():  
    return('Zinah')  
  
myname= getstring()      #  
print(myname)
```

↓ Output

'Zinah'

Parameters in functions

Write a function that take two number arguments and add 3 to first & 2 to the second and print the results

```
x=10  
y=10  
def increase(x,y):  
    x+=3  
    y+=2  
    print("x,y inside increase func.",x,y)  
increase(x,y)  
print("x,y in main program",x,y)
```

↓ Output

x,y inside increase func. 13 12
x,y in main program 10 10

The original values were not changed

Parameters in functions

Write a function that take a list as arguments and add 2 to each item and print the results

```
x=[1,2,3]
def increase(y):
    for i in range(len(y)):
        y[i]+=2
    print("y inside increase func.",y)
increase(x)
print("x in main program",x)
```

Output

```
y inside increase func. [3, 4, 5]
x in main program [3, 4, 5]
```

The original values were changed

Default Parameters in functions

Take number and exponent and print the result if the exponent exist otherwise print the square of this number

```
def mypower(x, y=2):
    R=x**y
    print(R)
```

```
mypower(2, 4)
mypower(2)
```

output

16
4

Default argument in functions

If the default argument not exist in the function call then it will take the default value

```
def printinfo( name, age = 35 ):
    "This prints a passed info into this function"
    print "Name: ", name;
    print "Age ", age;
    return;

printinfo( age=50, name="miki" );
printinfo( name="miki" );
```

Default argument

Default Parameters in functions

The default parameter should be after normal parameters

```
def power(y=2, x):
    result=1
    for i in range(y):
        result*=x
    print(result)
power(2,3)
power(2)
```

Output

SyntaxError: non-default argument follows default argument

Variable-length Parameters in functions

The default parameter should be after normal parameters

```
def summation (*A) :  
    z=0  
    for a in A:  
        z+=a  
    return(z)  
x=10; y=20; z=30  
print(summation(x,y))  
print(summation(x,y,z))  
print(summation(x,x,y,z,z))
```



Output

```
30  
60  
100
```

Variable-length arguments in functions

- An asterisk (*) is placed before the variable name that will hold the values of all non keyword variable arguments.
- The general form is:

```
def functionname([formal_args,] *var_args_tuple ):  
    "function_docstring"  
    function_suite  
    return [expression]
```

- An example

```
def printinfo( arg1, *vartuple ):  
    "This prints a variable passed arguments"  
    print "Output is: "  
    print arg1  
    for var in vartuple:  
        print var  
    return;  
  
printinfo( 10 );  
printinfo( 70, 60, 50 );
```

Sequence of parameters in functions

- ☐ Normal parameters
- ☐ Default parameters
- ☐ Variable-length parameters

Order not matter
in this case only

```
def printinfo( name, age ):  
    "This prints a passed info into this function"  
    print "Name: ", name;  
    print "Age ", age;  
    return;  
  
printinfo( age=50, name="miki" );
```